PROJECT REPORT: FEATURE ENGINEERING, MODEL OPTIMIZATION & PERFORMANCE COMPARISON

Task 2 — Advanced Machine Learning Workflow

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1. Executive Summary

In Task 1, a baseline Linear Regression model was successfully implemented to introduce the foundational Machine Learning lifecycle. Moving beyond basic execution, Task 2 focuses on industrial engineering practices: feature engineering via scaling, training non-linear estimators, and executing rigorous, objective model comparison arrays to select an optimal system for production environments.

2. Methodology & Preprocessing Pipeline

To mirror real-world machine learning pipelines where models are iteratively refined using measurable statistics, the following sequence was implemented:

- **Data Acquisition:** The official California Housing Dataset was programmatically loaded directly from the `scikit-learn` ecosystem, containing structured housing attributes such as Median Income, House Age, Average Rooms, and geographic tracking details.
- **Feature Scaling:** Implemented a `StandardScaler` to perform Z-score normalization across all numeric input domains. This converts features into a uniform range, preventing columns with larger numeric magnitudes (such as population or income metrics) from disproportionately dominating optimization passes.
- **Data Partitioning:** Utilizing `train_test_split`, the processed matrix was split into an **80% training set** to optimize weight spaces, and a **20% testing set** reserved strictly as unseen data to check against over-memorization.

3. Evaluated Model Architecture Rationale

Three distinct algorithms were tested side-by-side to capture varying patterns within the structural housing space:

- **Linear Regression:** Served as our controlled parametric baseline system.
- **Ridge Regression (alpha = 1.0):** Introduced an L2 regularization penalty to bound structural variance and mitigate structural overfitting tendencies.
- **Decision Tree Regressor (max_depth = 5):** Chosen specifically to construct orthogonal, non-linear boundaries capable of mapping structural localized housing premiums.

4. Empirical Evaluation & Performance Metrics

Following model training execution, predictive accuracy was calculated across Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the Coefficient of Determination (R^2 Score).

Model Performance Array Table

Model Variant	MAE	RMSE	R^2 Score
Linear Regression	0.533	0.746	0.576
Ridge Regression	0.533	0.746	0.576
Decision Tree (Max Depth = 5)	**0.522**	**0.724**	**0.600**

Key Analytical Observations

- **Linear vs. Regularized Linear Convergence:** Linear Regression and Ridge Regression converged with identical metrics down to three decimal places (R^2 = 0.576). This shows that at alpha = 1.0, the system feature mapping is already structurally stable, rendering minimal L2 co-linearity shrinkage.
- **Non-Linear Dominance:** The Decision Tree Regressor systematically outperformed both linear setups, driving the R^2 score up to **0.600** and suppressing error bounds to a lowest-in-class RMSE of **0.724** and MAE of **0.522**.

5. Engineering Conclusion & Model Selection Justification

Based strictly on measurable performance data, the **Decision Tree Regressor** is designated as the winning architecture for this deployment workflow.

While linear lines establish a reliable, high-speed baseline explaining 57.6% of pricing movements, real estate metrics depend inherently on complex geographic and density boundaries. The Decision Tree successfully intercepted these non-linear interactions, mapping an enhanced **60.0% of price variance**. Feature preprocessing via scaling successfully stabilized weight variations, ensuring reliable deployment readiness.